

SMART IRRIGATION SYSTEM :-

(Data Analysis Project Phases)

Phase 1: Problem Understanding

Understanding irrigation issues and water wastage. Goal is to analyze and improve water usage.

Phase 2: Data Collection & Loading

Load dataset using Python and Pandas.

Phase 3: Data Cleaning

Handle missing values, remove duplicates, and correct data types.

Phase 4: Data Understanding

Use info() and describe() to understand structure and statistics.

Phase 5: Exploratory Data Analysis (EDA)

Use graphs to analyze irrigation patterns, soil types, and environmental factors.

Phase 6: Insight Generation

Draw conclusions such as which soil needs more water and impact of rainfall.

Phase 7: Data Storage (MySQL)

Store cleaned data for easy access and management.

Phase 8: Dashboard Creation (Power BI)

Create visual dashboards for better understanding.

Phase 9: Simple Flask App (Optional)

Display analysis results through a web interface.

Phase 10: Conclusion & Recommendations

Provide suggestions to optimize irrigation and reduce water wastage.

Final Flow:

Data → Cleaning → Analysis → Visualization → Insights → Dashboard